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PROJECT REPORT

**Claim-Level Hallucination Detection in Retrieval-Augmented
Generation Systems: An Entailment-Based Verification Module**

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CANDIDATES' DECLARATION

We declare that this report, titled “Claim-Level Hallucination Detection in Retrieval-Augmented Generation Systems: An Entailment-Based Verification Module,” is a record of work carried out by us. It presents a problem formulation, a structured literature survey, a proposed system design, and an evaluation plan developed under the guidance of Dr. Usha Kiruthika S. All external sources consulted during the preparation of this report, including journal papers, benchmark documentation, and prior system designs, have been cited appropriately in the References section. This report has not been submitted, in whole or in part, to any other institute or university for the award of any degree or diploma.

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1. Introduction

Background: RAG and Hallucination

Retrieval-Augmented Generation (or RAG), the common shorthand for Retrieval-Augmented Generation, has established itself as the most prominent architectural pattern for grounding large language models in verifiable data. Rather than depending solely on the parameters learned during pretraining, the RAG approach retrieves several passages related to a particular query from a knowledge base and conditions the generation process of a language model on the basis of the retrieved context. It seems obvious that, if one provides the model with the right sources, it will not need to invent anything since all the necessary facts are readily available in the input window of the language model. This is not quite true, however. A growing number of recent papers that focus on different RAG-related benchmarks (RAGTruth, HaluBench), span-level detectors (LettuceDetect), and system-level evaluators (RAGAS, ARES) prove that language models continue generating facts regardless of the passage provided to them, sometimes even contradicting their sources. This phenomenon is called hallucination in RAG literature, and the fact that it persists in the state-of-the-art proprietary and open-weight models demonstrates why this architectural approach is insufficient: the retrieval process does not guarantee the faithful usage of provided evidence.

Stakes for High-Consequence Applications

The implication of this problem is clear and becomes increasingly alarming with the transition of RAG systems from search-assistant use-cases to higher-stakes tasks like clinical decision support, financial question answering, legal document review, and enterprise knowledge assistants. In all of these applications, an answer that sounds coherent and plausible, although it is not grounded in the passage retrieved by RAG, poses a much larger threat than any answer which is obviously wrong, as a fluent hallucination is harder to detect. In this sense, a person reviewing an AI-generated summary of patient records would not necessarily check every sentence from the passage to see whether it is true. The system would fail its purpose in case even one clause in an otherwise accurate paragraph contains a false dosage, a number, or a term from the contract. It is the main idea behind the current project.

Limitations of Answer-Level Evaluation

The fundamental problem is that current measures of RAG reliability assess faithfulness at the wrong scale. The common practice, illustrated both by commercial grounding scorers and by various consistency detectors like SelfCheckGPT, is to compute a single scalar faithfulness score per entire answer. While the answer-level perspective is helpful as a quick check, it does not answer the key question of the user of the system: where in this answer should I believe the claim

made, and where do I need to verify it myself? An answer consisting of five sentences, of which four are completely grounded in the evidence and only one contains a fabricated statistic, will usually score fairly well overall due to the four correctly grounded sentences overwhelming the one problematic sentence. The one erroneous sentence simply averages out.

Evidence of Underestimated Hallucinations

This is not a theoretical problem; numerous studies used in this project to analyze hallucination, including studies on span-level hallucination detection and hierarchical verification framework RT4CHART, confirm that answer-level scores underestimate the actual problem prevalence, and re-annotation of benchmarks using finer granularity reveals significantly more errors than the originally provided coarse labeling suggests. For example, re-annotation of RAGTruth performed by RT4CHART found more instances of hallucinations than was previously identified using span-level labeling. This indicates that coarse answer-level and even span-level benchmarks have underestimated the problem from the very beginning.

Limitations of Binary Localization

Another, closely related problem is that many of existing detectors, even though they attempt to localize errors, reduce their localization down to a binary choice of either hallucinated or not-hallucinated. This kind of localization loses information that is needed for the repair or verification processes. An utterance generated in a RAG pipeline can be in relation to the retrieved context in a few ways, not just in two, and equating every ungrounded utterance with others regardless of its nature, from being a completely absurd fabrication to a believable yet unverifiable aside, loses the precise distinction that the system designer needs in order to decide what action to take: reject the answer, mark it as questionable, or accept it with a note of warning.

Proposed Claim-Level Approach

This work directly tackles these two challenges by introducing a claim-level, entailment-based verification module for RAG pipelines. In essence, verification is treated not as one single classification problem at the level of answer, but rather as two steps. Firstly, the answer is decomposed into a set of independent facts claims, following the principle of atomic fact decomposition in various fact-checking evaluation frameworks like FActScore and fact-checking pipelines overall. Secondly, for each claim, an entailment check against relevant retrieved passages is done based on a natural language inference formulation that categorizes claims into one of three labels: supported, unsupported, or contradicted, similarly to the classical entailment, neutral, and contradiction categories used in NLI literature from early NLI datasets

like SNLI and MultiNLI up to more recent NLI-based hallucination detection works like DeBERTa fine-tuned on MNLI, FEVER, and ANLI.

Structured Claim-Level Reporting

Instead of getting just one opaque score, this results in a structured claim-level report where exactly which statements are justified, which ones were created by the model out of thin air, and which ones contradict the very sources that were retrieved by the system, can be identified. This approach is carefully crafted to align with the direction of development of the field concerning the evaluation of faithfulness in RAG systems. There exist systems such as RAGAS and ARES that already perform the decomposition of the answer into atomic claims and check each of them against context using an LLM judge. More recent claim-level frameworks like CLATTER additionally perform decomposition of individual claims into sub-claims and aggregation of entailment and attribution evidence from the bottom up until the final conclusion is made. Domain-specific implementations of the approach, such as MedRAGChecker for biomedical question answering and FinGround for financial claims verification, show that claim-level, entailment-grounded verification easily generalizes to any domain and is especially useful where a single wrong number or clinical statement could be extremely dangerous.

2. Literature Survey

S. No.	Year	Paper Title	Authors	Journal	Publisher	Study Type	Hallucination Focus	Method / What They Did	Key Finding	Highlighted Research Gap	Why It Matters for New Research	Indexing	Source URL
1	2024	Detecting hallucinations in large language models using semantic entropy	Sebastian Farquhar; Jannik Kossen; Lorenz Kuhn; Yarin Gal	Nature	Springer Nature	Hallucination detection	Confabulation / uncertainty	Introduces semantic entropy: clusters semantically equivalent answers and measures uncertainty over meanings rather than token sequences.	Semantic entropy can identify many confabulations without requiring a reference answer and improves over naïve uncertainty measures.	Primarily targets uncertainty-driven confabulations; does not cover all factual, citation, reasoning, tool-use, or long-context hallucinations. Cost of repeated sampling and calibration across domains/models remains open.	Strong baseline for uncertainty-aware detection; can be combined with evidence verification and agent-based checking.	Yes, SCIE, Q1	https://www.nature.com/articles/s41586-024-07421-0
2	2024	Quantifying the uncertainty of LLM hallucination spreading in complex adaptive social networks	Guozhi Hao; Jun Wu; Qianqian Pan; Rosario Morello	Scientific Reports	Springer Nature	Hallucination propagation	Social-network spread of false LLM content	Models how hallucinated LLM content can propagate through complex adaptive social networks and quantifies uncertainty in spreading processes.	Shows that hallucination risk is not only a model-level issue; network structure and interaction dynamics can amplify misinformation.	Focuses on propagation after hallucination occurs, not on detecting/correcting the original claim. Needs validation on real multi-platform, multilingual, time-evolving networks and modern agent ecosystems.	Useful for studying downstream impact and risk-aware deployment rather than only model accuracy.	Yes, SCIE, Q1	https://www.nature.com/articles/s41598-024-66708-4
3	2024	Woodpecker: hallucination correction for multimodal large language models	Shukang Yin; Chaoyou Fu; Sirui Zhao; et al.	Science China Information Sciences	Springer Nature	Hallucination correction	Multimodal / visual hallucination	Uses a post-hoc correction pipeline that identifies questionable visual claims, gathers evidence, and revises the generated answer.	Demonstrates that external evidence and structured correction can reduce object-level hallucinations in multimodal LLM outputs.	Pipeline is relatively complex and dependent on auxiliary models/tools; correction is not a formal guarantee. Generalization to non-visual LLM hallucinations, video, long documents, and real-time agents remains limited.	Important example of detect→verify→correct rather than detection alone.	Yes, SCIE, Q1	https://link.springer.com/article/10.1007/s11432-024-4251-x
4	2024	Towards trustworthy LLMs: a review on debiasing and dehallucinating in large language models	Z. Lin; et al.	Artificial Intelligence Review	Springer Nature	Review	Bias + hallucination taxonomy and mitigation	Reviews causes, evaluation, debiasing, hallucination detection and mitigation strategies for trustworthy LLMs.	Organizes a rapidly growing field and highlights links between trustworthiness dimensions rather than treating hallucination in isolation.	Review-level synthesis; limited empirical head-to-head validation of methods. Rapid model evolution makes static taxonomies/benchmarks quickly outdated. More standardized, reproducible, multilingual evaluation is needed.	Good foundation for defining categories and positioning a new framework.	Yes, SCIE, Q1	https://link.springer.com/article/10.1007/s10462-024-10896-y

S. N o.	Year	Paper Title	Authors	Journal	Publisher	Study Type	Hallucination Focus	Method / What They Did	Key Finding	Highlighted Research Gap	Why It Matters for New Research	Indexing	Source URL
5	2025	A framework to assess clinical safety and hallucination rates of LLMs for medical text summarisation	E. Asgari; et al.	npj Digital Medicine	Springer Nature	Clinical evaluation framework	Medical summarisation / hallucination / omission	Develops a framework to quantify hallucinations, omissions and clinically important safety errors in generated medical summaries.	Shows that conventional text-similarity metrics are insufficient for clinical safety and that clinically weighted evaluation is necessary.	Domain- and task-specific; requires expert annotation and may not transfer directly to open-domain, reasoning, RAG, or agentic settings. Automated severity-aware verification remains underdeveloped.	Strong model for severity-weighted hallucination evaluation in high-stakes domains.	Yes, SCIE, Q1	https://www.nature.com/articles/s41746-025-01670-7
6	2025	A hallucination detection and mitigation framework for large language model-based text summarization	S. Liu; et al.	Scientific Reports	Springer Nature	Detection + mitigation	Text summarisation / factuality	Combines an LLM-based hallucination detector with iterative modification to revise hallucinated summaries across benchmark datasets.	Iterative detect-and-revise improves factual consistency over single-pass generation.	Still relies on LLM judgement, which can itself hallucinate; lacks formal evidence entailment and broad testing across domains, long context, multilingual data, and newer reasoning models.	Directly motivates a closed-loop detector + repair + independent verification architecture.	Yes, SCIE, Q1	https://www.nature.com/articles/s41598-025-31075-1
7	2025	Mitigating spatial hallucination in large language models via graph-based reasoning and value-guided planning	H. Zhang; et al.	Scientific Reports	Springer Nature	Reasoning mitigation	Spatial / long-horizon reasoning hallucination	Extracts graph structure from textual spatial problems and introduces value-guided reasoning to reduce inconsistent planning and dead-end errors.	Structured representations improve spatial reasoning reliability compared with unstructured prompting.	Limited to structured spatial tasks; unclear transfer to factual hallucination, open-domain QA, code, scientific reasoning and noisy real-world inputs. Dependence on correct graph extraction is a failure point.	Supports neuro-symbolic/graph-grounded approaches to hallucination reduction.	Yes, SCIE, Q1	https://www.nature.com/articles/s41598-025-93601-5
8	2025	User-reported LLM hallucinations in AI mobile apps reviews	R. Massenon; et al.	Scientific Reports	Springer Nature	Real-world observational study	User-perceived hallucinations	Analyzes user reviews of AI mobile applications to identify how hallucinations are experienced and reported outside benchmark settings.	Provides real-world evidence that hallucination is a user-facing reliability problem, not only a laboratory metric.	User reports are noisy, subjective and incomplete; cannot directly identify root cause or true hallucination rate. Needs linkage between user complaint, prompt, model version, context and verified ground truth.	Useful for creating human-centered evaluation and deployment-risk metrics.	Yes, SCIE, Q1	https://www.nature.com/articles/s41598-025-15416-8
9	2025	A scalable framework for evaluating multiple language models with hallucination identification and factual verification	S. Chakraborty; et al.	Scientific Reports	Springer Nature	Multi-model evaluation	Factual verification / hallucination identification	Builds a retrieval-supported framework comparing responses from multiple LLMs using semantic similarity, TF-IDF, NER and factuality-related checks.	Demonstrates practical multi-model comparison and retrieval-based support for identifying questionable responses.	Heuristic similarity and NER signals do not establish entailment or truth. Needs claim-level evidence attribution, calibrated uncertainty, adversarial evaluation and independent verifier validation.	Relevant to ensemble and multi-agent verification systems.	Yes, SCIE, Q1	https://www.nature.com/articles/s41598-025-15203-5

S. N o.	Year	Paper Title	Authors	Journal	Publisher	Study Type	Hallucination Focus	Method / What They Did	Key Finding	Highlighted Research Gap	Why It Matters for New Research	Indexing	Source URL
10	2025	HaluCheck: Explainable and verifiable automation for detecting hallucinations in LLM responses	Sangwoo Heo; Sungwook Son; Hyunwoo Park	Expert Systems with Applications	Elsevier	Explainable hallucination detection	Response-level factuality	Introduces a visualization and decision-support system that estimates hallucination likelihood and exposes evidence to users.	Emphasizes explainability and user-facing verification rather than only returning a binary hallucination label.	User-centered visualization does not itself guarantee factual correctness. Needs stronger claim-level grounding, cross-domain robustness, calibration, longitudinal evaluation and automated correction.	Good reference for explainable hallucination auditing interfaces.	Yes, SCIE, Q1	https://www.sciencedirect.com/science/article/pii/S0957417425003343
11	2025	Mitigating reasoning hallucination through Multi-agent Collaborative Filtering	Jinxin Shi; Jiabao Zhao; Xingjiao Wu; Ruyi Xu; Yuan Hao Jiang; Liang He	Expert Systems with Applications	Elsevier	Multi-agent mitigation	Reasoning hallucination	Uses multiple LLM agents to generate diverse reasoning paths and collaborative filtering to select more reliable answers.	Multi-agent diversity can reduce reasoning hallucination and improve answer reliability compared with a single reasoning path.	Agents may share correlated errors and increase cost/latency. Voting/filtering does not prove correctness; performance under adversarial, domain-specific and tool-using settings needs deeper study.	Highly relevant for extending to specialized verifier agents and closed-loop repair.	Yes, SCIE, Q1	https://www.sciencedirect.com/science/article/abs/pii/S0957417424025909
12	2025	Hallucinations of large multimodal models: Problem and countermeasures	Shiliang Sun; Zhilin Lin; Xuhan Wu	Information Fusion	Elsevier	Review / framework	Multimodal hallucination	Analyzes causes and categories of hallucination in large multimodal models and organizes mitigation strategies.	Highlights that hallucination emerges from both language-generation behavior and cross-modal misalignment.	Needs standardized multimodal benchmarks, temporal/video hallucination evaluation, causal diagnosis and unified metrics that compare modalities and model families.	Useful for extending LLM hallucination research beyond text-only systems.	Yes, SCIE, Q1	https://www.sciencedirect.com/science/article/abs/pii/S1566253525000430
13	2026	HaluGNN: Hallucination detection in large language models using graph neural network	Linggang Kong; Yunlong Zhang; Xiaofeng Zhong; Haoran Fu; Yongjie Wang; Huijun Liu	Expert Systems with Applications	Elsevier	Hallucination detection	Internal-state / graph-based detection	Represents relationships in LLM internal information as graphs and applies GNN-based classification for hallucination detection.	Shows graph structure can capture relational patterns missed by simple static hidden-state classifiers.	Requires white-box/internal access and task-specific training; robustness across architectures and model updates is uncertain. Causal rather than correlational graph explanations remain a gap.	Strong candidate for hybrid internal-state + evidence-based detection.	Yes, SCIE, Q1	https://www.sciencedirect.com/science/article/abs/pii/S0957417425044720
14	2026	Large language modeling of hallucinatory problem mitigation based on the wheel of emotions	Zhenyu Wang; Jianmin Wang; Zenan Lu; Fang You	Neural Networks	Elsevier	Hallucination mitigation	Generation control / emotion-inspired representation	Proposes a wheel-of-emotions-inspired strategy to model and mitigate hallucinatory generation behavior.	Reports improved mitigation performance through structured control of generation signals.	The conceptual mechanism is specialized and needs independent replication, broader benchmark coverage, interpretability analysis and comparison against modern RAG/verifier/self-correction baselines.	Useful as an alternative mitigation family beyond retrieval and simple self-checking.	Yes, SCIE, Q1	https://www.sciencedirect.com/science/article/abs/pii/S0893608025008779

S. N o.	Year	Paper Title	Authors	Journal	Publisher	Study Type	Hallucination Focus	Method / What They Did	Key Finding	Highlighted Research Gap	Why It Matters for New Research	Indexing	Source URL
15	2026	Hallucination to truth: a review of fact-checking and factuality evaluation in large language models	S. S. Rahman; M. A. Islam; M. M. Alam; et al.	Artificial Intelligence Review	Springer Nature	Review	Fact-checking / factuality evaluation	Synthesizes LLM factuality evaluation, fact-checking, retrieval, verification and hallucination mitigation literature.	Connects hallucination research with the broader fact-checking pipeline and evaluation ecosystem.	Field still lacks universal claim-level metrics, consistent benchmark definitions, robust multilingual evaluation and reliable verifier calibration. Reviews cannot establish which combined architecture works best in deployment.	Excellent foundation for identifying open gaps in evidence-grounded verification.	Yes, SCIE, Q1	https://link.springer.com/article/10.1007/s10462-025-11454-w
16	2026	Large language models hallucination: A comprehensive survey	Aisha Alansari; Hamzah Luqman	Computer Science Review	Elsevier	Comprehensive survey	Causes, detection, metrics, mitigation	Provides a lifecycle-oriented taxonomy of causes, detection approaches, mitigation strategies, benchmarks and evaluation metrics.	Offers a current consolidated map of hallucination research across the LLM development and inference lifecycle.	Taxonomy is broad but mostly synthesizes existing methods; rapidly changing models and agentic/RAG systems require continuously updated empirical benchmarks and standardized evaluations.	Best recent survey for framing a thesis/paper literature review.	Yes, SCIE, Q1	https://www.sciencedirect.com/science/article/abs/pii/S157401372600078X
17	2026	Evaluating large language models for accuracy and hallucination	A. T. Kalai; et al.	Nature	Springer Nature	Theory + evaluation	Why models hallucinate / guessing incentives	Uses learning-theoretic analysis and evaluation studies to show that next-token prediction and common accuracy benchmarks can incentivize guessing rather than calibrated abstention.	Argues that hallucination is partly driven by training/evaluation incentives and that models should be rewarded for acknowledging uncertainty.	Does not by itself provide a complete production-ready detector/repair system. Open problems include calibrated abstention under distribution shift, tool use, long context and domain-specific risk thresholds.	Important theoretical basis for uncertainty-aware abstention mechanisms.	Yes, SCIE, Q1	https://www.nature.com/articles/s41586-026-10549-w
18	2026	Hyper-RAG: combating LLM hallucinations using hypergraph-driven retrieval-augmented generation	Yifan Feng; Hao Hu; Shihui Ying; Xingliang Hou; Shiquan Liu; et al.	Nature Communications	Springer Nature	RAG mitigation	Knowledge-grounded hallucination	Uses hypergraph-based retrieval to model higher-order relations among domain knowledge and improve grounding before generation.	Reports lower hallucination/error propagation and better knowledge coverage than standard graph/RAG baselines across multiple datasets.	Dependent on knowledge-base quality and retrieval construction; does not eliminate generation-side reasoning errors. Dynamic knowledge, conflicting evidence, provenance quality, and adversarial retrieval remain open.	Strong recent RAG baseline for domain-specialized hallucination mitigation.	Yes, SCIE, Q1	https://www.nature.com/articles/s41467-026-71411-1
19	2026	Mitigating hallucinations in healthcare LLMs with granular fact-checking and domain-specific adaptation	Musarrat Zeba; Abdullah Al Mamun; Kishoar Jahan Tithee; et al.	Expert Systems with Applications	Elsevier	Fact-checking + fine-tuning	Healthcare summarisation	Combines LoRA-based domain adaptation with an LLM-independent granular fact-checking module using numerical and logical validation against EHR facts.	Shows strong proposition-level fact-checking performance and demonstrates value of separating verifier logic from the generator.	Evaluated mainly on MIMIC-III summaries; external clinical validation, temporal data, multilingual records, reasoning-heavy tasks and formal safety	Very useful model for independent verification rather than self-checking by the same LLM.	Yes, SCIE, Q1	https://www.sciencedirect.com/science/article/pii/S0957417426018786

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										guarantees are still missing.			
20	2026	A framework for hallucination mitigation in domain-specialized large language models with application to aviation maintenance decision support	Xuanting Lu; et al.	Information Processing & Management	Elsevier	Domain-specific mitigation	Aviation maintenance decision support	Develops a hallucination-mitigation framework for specialized LLM decision support in aviation maintenance, emphasizing domain grounding and reliability.	Demonstrates the importance of domain-specific knowledge and validation for high-stakes industrial LLM deployment.	Evidence is concentrated in one industrial domain; needs cross-industry replication, standardized severity metrics, real operational trials, uncertainty calibration and integration with formal/causal verification.	Strong industrial example for testing whether a general hallucination framework transfers across domains.	Yes, SCIE, Q1	https://www.sciencedirect.com/science/article/abs/pii/S0306457326001536

Table 2.1: Summary of twenty peer-reviewed Q1/SCIE/Scopus-indexed journal papers (2024–2026) surveyed for this report.

Overview of the Literature

The literature consulted in this project is based on twenty peer-reviewed articles in journals with SCIE/Q1 ratings and indexed by Scopus and Web of Science, covering publications between 2024 and 2026 by various publishers, such as Nature, Springer, and Elsevier, in addition to a more focused collection of recent articles on the topic of claim-level and entailment-based verification of RAG. When viewed collectively, these sources form a coherent narrative, from initial uncertainty-based detection, through answer-level fidelity scoring, and ending up with the claim-based verification against an evidential basis used in the current project.

Uncertainty-Based Detection

The first line starts with detecting hallucination in terms of certain signals inside the model generation process itself, where there is no external information to be considered. The semantic entropy method introduced by Farquhar et al. (Nature) classifies semantically equivalent sampled answers into groups, measures the uncertainty over meaning, and detects many confabulations, not requiring ground truth. SelfCheckGPT takes this approach further as a practical black-box detector: it samples multiple responses to the same prompt and identifies the statements that are inconsistent among them as hallucinations.

Limitation of Self-Consistency

These methods are very useful exactly because they do not require a retrieval corpus at all, but have one important limitation that is related to this project – consistency among samples is not equal to truth. A model may provide consistent but incorrect answers, and the fact of self-consistency does not guarantee correctness at all. That is one of the reasons why the current project is focused on evidential verification.

Retrieval-Side Enhancements

A second thread focuses on moving from unconditional generation to retrieval augmentation specifically, and investigates whether retrieval alone is indeed enough to solve the grounding problem. Hyper-RAG, an article published in Nature Communications, uses hypergraphs for evidence retrieval to represent higher-order relationships between fragments of domain knowledge and reports lower levels of hallucination and error propagation compared to traditional graph- and vector-based RAG baselines. This paper is part of a family of works focusing on retrieval-side enhancements surveyed as part of this project, all of which aim to provide more consistent evidence to the generator.

The Limits of Better Retrieval

The important thing about these works, however, is the fact that none of the surveyed papers focused solely on retrieval state that an enhanced architecture will be sufficient to eliminate the

problem of hallucination. Generation-side reasoning failures, where the model misinterprets or fails to incorporate correctly retrieved evidence, are left untouched. Dynamic knowledge, conflicting evidence across multiple sources, and provenance quality are pointed out explicitly as open issues, even when the most sophisticated retrieval architectures are considered. In other words, while better retrieval reduces the surface area for hallucination, it does not bridge the gap between the correctly retrieved evidence and the generated output.

Need for Independent Verification

It is exactly this separation between retrieval quality and generation faithfulness that constitutes the underlying assumption of the present project design. Verification of the generated output against the retrieved passages, carried out in an independent layer, is required because it does not matter what retrieval quality the system uses; the only thing that matters is making sure the generator did use it. Even a system using a near-perfect hypergraph-based retriever still requires a means to make sure that the generator did not paraphrase or extrapolate upon the retrieved evidence, but used it faithfully.

System-Level Faithfulness Scoring

A third line of work involves the predominant evaluation methodology used today to measure the faithfulness of RAG systems: system-level scoring schemes such as RAGAS and ARES. In RAGAS, the generated answer is divided into atomic claims and each claim is compared against the retrieved context using an LLM judge, with the total faithfulness score being calculated as a fraction of claims that were found to be supported. ARES, in turn, trains lightweight classifiers that are able to score RAG systems without the need to exhaustively annotate for every new deployment. Both these methods, while representing an improvement in comparison to just answer-level scalar scoring, still perform claim decomposition internally before aggregation, yet the final output presented to the user is, in most cases, a single aggregated faithfulness score per answer or per RAG run.

Limitations of Aggregated Scores

And here we encounter the exact same problem as described in Section 1. The faithfulness scores measured by RAGAS in various benchmarking studies surveyed for this project fall in a moderate range: some evaluations report mean faithfulness between 0.5 and 0.9 depending on the domain and retrieval method, which tells you something about the overall reliability of a system, but does not provide any information on whether or not you can trust a particular claim within a particular answer. In one benchmarking study, it is explicitly noted that faithfulness scores under a no-context ablation tend to hover near perfect, as the metric measures the lack of contradictions and there is none to detect when no context is provided at all.

Preserving Individual Claim Verdicts

The decomposition of claims performed by both RAGAS and ARES is valuable in and of itself and informs the construction of the module that is the core of this project. This project changes the step that follows decomposition: instead of re-aggregating the per-claim judgments into one number, the individual three-way classification of each claim is preserved and presented.

Span-Level and Hierarchical Detection

A fourth strand ranges from the approach of answer-level and system-level scoring into span-level and hierarchical detection; this design is close to the one used in this project. LettuceDetect exemplifies this approach: it does not give a single yes/no score for the generated response but highlights problematic parts of it, though it does not provide any corresponding evidence in the context explaining why that particular span has been marked as problematic, making this tool less useful for verification. RT4CHART goes even further by adding a hierarchical two-pass design: local verification is performed on overlapping context chunks, and then global verification is run again on each claim with the help of local results. As a result, it gives one of three labels — entailed, contradicted, or baseless — along with the span containing the contradicting or supporting evidence.

Importance of Fine-Grained Annotation

RT4CHART’s evaluation revealed a piece of information highly relevant to the motivation behind this project: when the authors re-annotated the RAGTruth benchmark at a more granular level, they found 1.68 times more hallucination cases than had been identified in the original annotation. This is strong empirical proof that benchmarks designed explicitly for detecting hallucinations tend to be incomplete in counting such cases due to their initial coarse annotation.

The three-label scheme used in RT4CHART — entailed, contradicted, baseless — is very similar to the one proposed for the verification module in this project — supported, unsupported, and contradicted — and the results show that this three-way division, rather than a binary one, is the latest trend in the span-level approach to verification.

Encoder-Based Entailment Models

The fifth theme relates to the actual computational feasibility of claim-level verification through the technology of entailment: encoder-based natural language inference (NLI) models. Natural language inference has a lengthy history of research, centered on the classification of natural-language sentences into one of three canonical classes — entailment, neutral, and contradiction — pioneered by the SNLI dataset and extended by adversarial and fact-verification-oriented datasets such as ANLI and FEVER. For the specific task of hallucination detection, the model architecture that has become ubiquitous is an encoder fine-tuned jointly on MNLI, FEVER, and

ANLI, used across the surveyed papers as HHEM-type detectors and repurposed in evaluation suites such as VERITAS, which shows that encoder-based entailment models such as DeBERTa outperform even generative reasoning models of equivalent or larger size on the task of claim verification, because entailment is a narrower and structurally simpler task than open-ended generation, and thus better suited to direct training of encoders for classification.

Sub-Claim Decomposition

CLATTER is an advancement on this idea of entailment, but goes one step further: while conventional fact verifiers test a claim against a source text in a single move, CLATTER first breaks down the claim into smaller sub-claims, tests the sub-claims separately against sources for attribution and entailment evidence, and only then makes a final accept/reject verdict based on their results. The decomposition of claims into sub-claims is a refinement that is recognized by the design of the current project but not fully taken advantage of in its first iteration due to considerations of efficiency outlined in Section 3; however, it is noted explicitly as a potential improvement.

Domain-Specific Verification

The sixth line of research, again taken almost entirely from the entailment-verification literature, involves domain-specific applications of the technique that attempt to test whether verification at the claim level using entailment works in domains other than open-domain QA systems. MedRAGChecker tests claim-level verification in biomedical RAG, breaking down complex answers into their constituent claims and classifying each one as either entailed, contradicted, or missing reference. The reasoning here is that scoring the entire answer as one unit would not necessarily reveal problems with claims hidden within an otherwise well-justified response. The paper adds an additional layer by considering a knowledge-graph signal along with the entailment check to gauge both the validity of a claim and the amount of evidence supporting it, moving beyond what could be established through NLI modeling alone. FinGround applies atomic claim verification in financial question-answering systems, using it to analyze claims that involve numbers, such as gross margin percentage claims. It shows that generic claim decomposition treats all facts equally and misses many computational errors in the process.

Transferability and Domain Adaptation

Both projects reach the same conclusion relevant to this project: the architecture of decomposing a claim and verifying it is easily transferable between domains; however, what constitutes an atomic, checkable claim and the kind of evidence needed for verification depend on the domain. Both clinical and financial atomic claims involving numbers (e.g., dosages in clinical claims and percentages in financial claims) require exact value matching and may not be

weighted strongly enough in purely semantic entailment models. Since this project attempts to build a general-purpose RAG verification module, it seems reasonable to conclude that if this project eventually needs to adapt its module to a domain-specific application, it will be necessary to adapt the claim extraction part, rather than the entailment classification part.

Research Gaps

In synthesizing across all twenty papers and the entailment literature discussed above, there are a handful of recurring gaps that consistently appear to be worth pursuing for this project. The most frequently mentioned gap is the lack of a unified claim-level verification standard: most systems currently in place use uncertainty scores, similarity heuristics, or even the opinion of another LLM as a judge to verify claims made by a system, but none use a unified standard for identifying the exact false claim, connecting it to the particular evidence that contradicts or supports it, and explaining the decision-making process in interpretable terms. Another recurrent gap is that there has been more work done on detecting claims than repairing them, that is, while current systems can flag certain claims and either refine them or fix them iteratively, no system exists where detection is followed by deterministic verification with an accept-or-reject certificate before the answer is delivered to users. A further recurrent gap, most clearly articulated in the review on debiasing and dehallucinating LLMs, is that self-verification is inherently structurally unreliable in these cases, because the judge LLM uses the same faulty generation mechanism as the generator, making a verification method independent of large language models necessary. Finally, another recurrent gap, particularly evident in RAG-related papers including Hyper-RAG, is that while better retrieval systems reduce hallucination, it still persists, with the question remaining whether the particular claim made is entailed by the retrieved evidence.

In total, these four gaps constitute exactly the project space described in this proposal: a claim-level, evidence-grounded, three-way entailment verdict created using a deterministic classifier, independent of other LLMs acting as judges.

3. Problem Statement

By contrast, and precisely what is less studied and addressed in this project, is the problem of developing and implementing a lightweight encoder-based entailment verification component that can be integrated directly into the RAG serving pipeline in order to provide deterministic structured claim verdicts in real time without making an additional expensive and potentially self-hallucinating generation model call. Such a solution would be cheaper and more reliable

compared to LLM-as-judge verification and would not require the use of an additional generative model at all. To restate the problem in this way makes sense operationally. In particular, LLM-as-judge verification, although quite flexible, is costly, slow, and is prone to hallucinations, which is clearly noted in several of the analyzed papers including the review on mitigating hallucination and dehallowinating models. Specifically, LLM self-verification is explicitly pointed out as an unreliable and dangerous technique since the judge itself can hallucinate just as the judged system.

Thus, the problem statement for this project can be clearly formulated. Current RAG systems are monitored using measures of answer- or system-level faithfulness that are unable to localize the problematic part of an answer, and the approaches that make such localization possible either rely on binary support judgments or costly self-referential LLM-as-judge verification, which is itself an unreliable and non-deterministic verification process. There is a gap that has been repeatedly documented, both in the academic literature studied for this project and in current tooling, regarding the lack of a verification approach that combines the decomposition of generated answers into claims; the classification of each claim against the retrieved context using an explicit entailment relation of supported, unsupported, or contradicted rather than a single score; and an efficient, deterministic implementation of this classification procedure so it can be integrated as a natural part of the production RAG pipeline rather than as an offline audit process. This project aims to develop, and in a subsequent stage implement and evaluate, such a verification module, placing it explicitly in the context of current hallucination detection research in regard to the research gaps identified in the survey of twenty peer-reviewed Q1 journal papers published from 2024 to 2026, which forms the basis of the survey presented above.

4. Proposed System

4.1 Overview

The proposed system consists of three stages which operate after a regular RAG generator and before delivering the final answer to the user: claim decomposition, retrieval alignment, and entailment classification. The claim decomposition stage operates on the output of the RAG generator and decomposes the answer into a set of atomic claims, each expressing a single checkable fact, rather than multiple facts at once in a compound sentence. This stage is necessary due to the fact that entailment classification becomes unreliable when the hypothesis is not atomic: a compound sentence containing a grounded clause and an imagined clause cannot be reliably classified by an entailment model, because the model cannot distinguish the fact that

only half of the sentence is true while the other half is false. Following the approach of FActScore and RAGAS, the decomposition can be done either by using a large language model prompted to generate a list of atomic factual statements out of the input sentence, or by using a rule-based sentence-splitting and coreference-resolution pipeline to keep latency low, since an additional LLM call for each answer is not always desirable.

Retrieval Alignment

The second stage of the pipeline solves a very subtle problem: the extracted claim should be checked against the proper subset of the passages retrieved from the database, rather than against the entire retrieved context. Checking a claim against irrelevant passages consumes computational resources and introduces noise into the decision of the entailment model, which might fail to identify the actual sentence to compare against when the premise is lengthy and only partly relevant. This project solves the alignment problem via semantic similarity search between each claim and the retrieved passages, narrowing the set of premises passed to the entailment model down to the most relevant two or three passages per claim.

4.2 Entailment Classification

Three-Way Entailment Verdicts

The third and key stage is the entailment classification. For every claim paired with its filtered set of aligned passages, an NLI classifier outputs exactly one out of three possible verdicts: supported, if the aligned passages provide evidence that supports the claim; unsupported, if the aligned passages neither support nor contradict the claim, implying that the model made use of parametric knowledge not provided in the evidence; or contradicted, if the aligned passages provide evidence that directly contradicts the claim. For the purposes of this work, the recommendation is to implement the entailment classification module as an encoder-based NLI model, specifically a DeBERTa-family model fine-tuned on the MNLI, FEVER, and ANLI datasets, in line with the HHEM-style verifiers mentioned multiple times in the literature survey above. This decision is grounded in the research gap outlined in Section 2: encoder-based entailment classifiers are deterministic and cheap to compute at scale, and are fundamentally immune to the self-referential hallucination inherent to prompted generative LLM-judge verification.

Structured Verification Output

In addition to the claim-level verdict, the claim text, the passage span responsible for the decision, and a confidence score calculated from the output logits of the classifier are stored in the output record, meaning that the entailment classification produces not only a decision about whether the claim is true or false, but a structured set of information about the nature of that decision. These records are then aggregated into an answer-level report, preserving individual

verdicts rather than a summary value, thus ensuring that a downstream system or human reviewer can correctly interpret the results and know exactly which claims can be trusted and which should be rejected.

4.3 Aggregation Policy and Worked Example

Formal Aggregation Policy

In addition to providing verdicts on individual claim veracity, the module specifies a formal aggregation policy that dictates the reaction of the surrounding RAG system based on the classifications of all claims within an answer. Three thresholds determine the policy: the answer is left untouched if every claim is marked as supported; an answer with at least one unsupported but no contradicted claims is passed through, with those unsupported claims flagged in-line for the user, since an unsupported claim can still be true but cannot be backed up by the retrieved evidence; and an answer with any contradicted claims is either withheld from output or re-generated, since the presence of a contradiction indicates that the model made a statement which goes against facts stated by the system's own retrieved information. This three-tiered response is the direct result of maintaining the three-way distinction argued for in Section 1, rather than collapsing everything not grounded in the evidence into the same binary failure.

Worked Example

To provide a concrete worked example, imagine a scenario where a RAG system retrieves a passage and the generator produces an answer, as follows:

- **Retrieved passage:** a company reported quarterly revenue of 4.2 billion rupees with a net profit margin of 11 percent.
- **Generated answer:** "The company announced quarterly revenue of 4.2 billion rupees, with profit margins increasing as a result of cutting costs in manufacturing."
- **Decomposition:** the answer is decomposed into two atomic claims: first, that quarterly revenue was 4.2 billion rupees; and second, that profit margins increased due to cost-cutting in manufacturing.
- **Retrieval alignment:** the first claim is identified as corresponding to the retrieved revenue passage, whereas the second corresponds to the passage containing the 11 percent margin figure.
- **Entailment classification:** the first claim is marked as supported, since the number matches exactly, whereas the second claim, despite being aligned with a retrieved

passage, is marked as unsupported, correctly isolating the fabricated causal explanation while leaving the accurate revenue figure untouched.

4.4 Implementation and Deployment Considerations

Modular Deployment Design

Implementation-wise, the module is built to be added as a final, single step in an existing RAG serving pipeline, without changing anything in the retriever or generator itself. The stages of claim decomposition, retrieval alignment, and entailment classification are each designed as independently swappable components, such that a deployment optimized for minimum latency may use a rule-based sentence splitter in place of the LLM-powered claim decomposer, while a deployment optimizing for verification quality may keep the LLM decomposer and implement the sub-claim decomposition refinement proposed in the CLATTER framework discussed in Section 2. This flexibility in architecture reflects the project’s core contribution: the three-way, evidence-grounded entailment formulation, not a particular model implementation, and the architecture is designed to accommodate the efficiency-versus-completeness trade-off differently for different deployments, whether a high-throughput consumer chatbot or a lower-throughput, higher-stakes clinical or financial assistant.

Efficiency Considerations

Latency and cost efficiency are treated as first-class concerns throughout the project, reflecting the critique of LLM-as-judge verification raised in Section 3. Because the entailment classifier requires only a single inference pass through an encoder, rather than a full generative call, and because the retrieval alignment stage narrows the premises for each claim beforehand, the expected cost of per-claim verification is considerably lower than that of an LLM-judge alternative, although the exact magnitude of this efficiency gain, along with the module’s precision and recall on the classification task, is left to the experimental evaluation plan described in Section 5, since no benchmarking results have yet been collected for this proposal-stage report.

5. Evaluation Plan and Experimental Design

5.1 Benchmarks

Benchmark Selection

Evaluation of a claim-level entailment verifier needs benchmarks which offer ground-truth labels at the claim or span level rather than at the answer level alone, since all the reasoning behind this project stems from the premise that answer-level labels hide precisely the mistakes that this

module is supposed to identify. RAGTruth is the benchmark of choice in this regard, as it supplies span-level hallucination annotations across question-answering, data-to-text, and summarization tasks generated by a number of underlying language models, while the re-annotated version of the same benchmark, RAGTruth-Enhance, produced as part of the RT4CHART project outlined in Section 2, offers an even more robust span-labeling dataset compared to the original release. HaluBench is an additional benchmark which combines samples from specialized domain sources, such as CovidQA, PubMedQA, DROP, and FinanceBench, and is especially relevant for verifying that the performance of the proposed module holds beyond generic open-domain question answering, extending into the domain-sensitivity territory of the MedRAGChecker and FinGround modules outlined in Section 2.

5.2 Metrics

Evaluation Metrics

The metric of choice for evaluating this system will be claim-level classification performance, to be reported in terms of precision, recall, and F1 score for each of the three target labels: supported, unsupported, and contradicted. Instead of reporting a single pooled accuracy metric, which would be misleading given expected class imbalance, these numbers will be calculated separately for each class, and in particular for the rarer but more operationally significant contradicted class. Building on metric design from the multi-agent auditing literature reviewed during preparation for this project, both macro- and micro-averaged F1 scores will be reported, since the former favors balance among all three labels while the latter reflects the actual label distribution present in the benchmark.

5.3 Baselines and Ablations

Baseline Comparisons

In order to put the proposed module into context in terms of performance, three baseline comparisons will be carried out. The first is a Single-Pass LLM-Judge baseline that asks a large language model to assess the faithfulness of a generated answer together with the full retrieved context in a single prompt, without breaking the answer into claims, thus replicating the naive answer-level approach criticized throughout this report. The second is a RAGAS-style baseline, where claim decomposition is followed by LLM-judge verification of each claim, but instead of a three-way verdict per claim, an overall aggregated faithfulness score is produced for the entire answer, thus replicating current standard industry practice as described in Section 2. The third is a binary-classification ablation of the proposed module itself, where the entailment classifier is limited to two labels, hallucinated and not hallucinated, merging the unsupported and

contradicted categories, thus directly testing the hypothesis that the three-way categorization this project proposes yields better results than the more common binary framing, rather than relying on theory alone.

Ablation Studies

Aside from the baseline comparisons, two ablation studies will be carried out to identify which specific architectural decisions are responsible for the module’s performance, following the same ablation methodology used in the reference multi-agent auditing framework studied during preparation for this project. The first will remove the retrieval alignment stage, giving the entailment classifier the full retrieved context for all claims instead of two to three passages, in order to test whether retrieval alignment genuinely increases classification accuracy rather than merely accelerating it. The second will replace the encoder-based DeBERTa entailment classifier with a prompted LLM-judge classifier while leaving the claim decomposition and retrieval alignment stages intact, thus isolating the specific benefit provided by using a deterministic encoder instead of a generative LLM judge as the final classification tool.

5.4 Human Evaluation and Latency Measurement

Human Interpretability Evaluation

Benchmark metrics alone are not enough for evaluating an interpretability-focused module, and thus human evaluation is a part of the planned procedure alongside automated benchmarking. A subset of answers marked with mixed ratings, answers that have at least one supported claim and one contradicted or unsupported claim, will be drawn from benchmark test sets and given to independent annotators together with the structured output of the module, including the claim in question, its label, and the corresponding evidence span. For each such answer, annotators will be asked two questions: does the assigned label agree with the evidence span given, and was this span sufficient for them to form a judgment without consulting the full retrieved context. While the first question represents the classification-agreement measure familiar from inter-annotator agreement studies for NLI datasets, the second question allows verification of the actual practical usefulness of the module’s evidence attributions, a claim this report has made repeatedly but which F1 scores alone cannot verify.

Latency Measurement

Latency and throughput will be measured separately as a systems-level concern, reported as average per-answer verification time and the module’s added latency relative to the baseline response time of the underlying RAG generator. This metric directly evaluates the efficiency argument made against the LLM-as-judge approach in Sections 3 and 4, and the two ablations described above will be timed under identical conditions, so that any accuracy gained from

retrieving full context or from switching to an LLM-judge classifier can be weighed directly against its latency cost.

5.5 Expected Outcomes

Expected Performance

Based on the patterns observed across the surveyed literature, several expected outcomes emerge, which have not yet been confirmed, as this report outlines the design and evaluation plan of the module and does not yet include any experimental results. The proposed module is expected to deliver claim-level F1 scores comparable with, and in the case of the contradicted class likely outperforming, the RAGAS-style and Single-Pass LLM-Judge baselines, relying on the established pattern in the literature of RT4CHART and CLATTER, where structured, hierarchical, or sub-claim-decomposed verification outperformed single-pass judgment across comparable benchmarks. It is expected that the binary-classification ablation will demonstrate inferior performance to the full three-way module specifically in terms of downstream aggregation quality, even if its raw hallucinated-versus-not accuracy is comparable, since binary classification cannot distinguish the urgency of a contradiction from a merely unattributed claim, a distinction this report has argued is operationally relevant independent of raw accuracy. On latency, the proposed module is expected to exhibit substantially lower verification time per answer compared with both baseline types, consistent with the well-established efficiency gap between a single encoder forward pass and a full generative call documented in the systems literature reviewed in Section 2.

Status of Expected Outcomes

It must be stated clearly that none of these expected outcomes have been measured at this point: they are hypotheses arising from the literature review, not results produced by this project. The next immediate phase of work after submission of this report is implementation of the three-stage pipeline outlined in Section 4, followed by testing of the full evaluation plan outlined above using the RAGTruth-Enhance and HaluBench datasets.

6. Conclusion

In conclusion, this report has begun with a statement of the existing problem, that RAG faithfulness evaluation today is too coarse-grained and dependent on self-referential judgments, and has continued with a systematic literature review highlighting the gaps this problem leaves open, to reach the actual design of a three-stage system and an evaluation scheme intended to honestly test this design against real benchmarks, instead of merely asserting its value. The next

steps of this project are implementation and evaluation, and the results of these steps will determine the success of the claim-level verification design proposed in this report.

Limitations

It is necessary to consider a number of limitations associated with this proposal. The reliability of the proposed module depends directly on the reliability of the claim decomposition stage, and the three-way entailment approach, while better than binary classification, does not account for partially supported claims. Finally, this report presents only a design and evaluation proposal; the expected results discussed in Section 5 represent hypotheses based on the surveyed literature, not measured results.

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